Chapter 3 notes

p.1 3.2

The auto covariance function $\gamma(k)$

$$\gamma(k) = Cov(X_t, X_{t+k})$$

$$= E(X_t X_{t+k}) - E(X_t)E(X_{t+k})$$

If $k=0$, then

$$\gamma(0) = E(X_t^2) - [E(X_t)]^2 = Var(X_t)$$

The auto correlation function (ACF) $\rho(k)$

$$\rho(k) = \frac{Cov(X_t, X_{t+k})}{\sqrt{Var(X_t)\,Var(X_{t+k})}} \quad \text{since } V(X_t) = V(X_{t+k})$$

$$= \frac{Cov(X_t, X_{t+k})}{V(X_t)} = \frac{\gamma(k)}{\gamma(0)}, \quad k = 0, \pm 1, \pm 2, \ldots$$

Note that $\rho(0) = 1$.

3.3.1 White noise

$$X_t = Z_t, \quad t = 0, \pm 1, \pm 2, \pm 3, \ldots$$

where $\{Z_t\} \overset{iid}{\sim} N(0, \sigma^2)$, i.e., $E(Z_t) = 0$, $V(Z_t) = \sigma^2$.

$$\gamma(k) = E(X_t X_{t-k}) = \begin{cases} \sigma^2, & k=0; \\ 0, & \text{otherwise.} \end{cases}$$

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \begin{cases} 1, & k=0; \\ 0, & k \neq 0. \end{cases}$$

3.3.2 AR(1)

$X_t = \alpha X_{t-1} + Z_t$, where $|\alpha| < 1$, $\{Z_t\} \overset{iid}{\sim} N(0, \sigma^2)$

and Z_t and X_{t-k} are uncorrelated, $k \neq 0$.

$$E(X_t) = E[\alpha X_{t-1} + Z_t] = \alpha E(X_{t-1}) + E(Z_t)$$

$$\mu = \alpha\mu + 0$$

$$\Rightarrow \mu = 0$$

$$V(X_t) = V[\alpha X_{t-1} + Z_t] = \alpha^2 V(X_{t-1}) + V(Z_t) + 2\,Cov(\alpha X_{t-1}, Z_t)$$

$$= \alpha^2 V(X_t) + \sigma^2 + 0$$

$$\Rightarrow V(X_t) - \alpha^2 V(X_t) = \sigma^2$$

$$V(X_t)(1 - \alpha^2) = \sigma^2$$

$$V(X_t) = \frac{\sigma^2}{1-\alpha^2}$$

$$\gamma(k) = E(X_t X_{t-k}) = E[(\alpha X_{t-1} + Z_t) X_{t-k}]$$

$$= E(\alpha X_{t-1} X_{t-k} + Z_t X_{t-k})$$

$$= \alpha E(X_{t-1} X_{t-k}) + E(Z_t) E(X_{t-k})$$

$$= \alpha \gamma(k-1) + 0$$

$$\frac{\gamma(k)}{\gamma(0)} = \frac{\alpha\gamma(k-1)}{\gamma(0)} \Rightarrow \rho(k) = \alpha\rho(k-1)$$

$$\rho(k-1) = \alpha\rho(k-2)$$

$$\vdots$$

$$\rho(1) = \alpha\rho(0) = \alpha \qquad \text{N.B. } \rho(0) = 1.$$

$$\therefore \rho(k) = \alpha^k \rho(0) = \alpha^k.$$

Theorem 3.1 : AR(1) : $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$.

$$X_t - \phi X_{t-1} = \varepsilon_t \Rightarrow (1-\phi B) X_t = \varepsilon_t$$

$$X_t = (1-\phi B)^{-1} \varepsilon_t = (1+\phi B + \phi^2 B^2 + \cdots)\varepsilon_t$$

$$= \varepsilon_t + \phi B \varepsilon_t + \phi^2 B^2 \varepsilon_t + \cdots = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} \cdots = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \quad \text{(particular solution)}$$

(3.2)

and $X_t - \phi X_{t-1} = 0$

$$\Rightarrow X_t = \phi X_{t-1} = \phi^2 X_{t-2} = \phi^3 X_{t-3} = \cdots = \phi^t X_0$$

For initial condition, say, $X_0 = k$, then

$$X_t = k\phi^t \quad \text{(complementary solution)}$$

Thus the general solution of AR(1) is

$$X_t = k\phi^t + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \qquad (3.3)$$

As $t \to \infty$, then $k\phi^t \to 0$, hence

$$X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$$

p.3 Thus $E(X_t) = E\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}\right) = 0$, independent of t.

and $V(X_t) = \dfrac{\sigma^2}{1-\phi^2}$. (see page (2) of AR(1) with $\alpha = \phi$)

and $Cov(X_t, X_{t-k}) = E(X_t X_{t-k}) = \gamma(k)$

$$= \phi \gamma(k-1) = \phi^2 \gamma(k-2) = \cdots = \phi^k \gamma(0)$$

$$= \phi^k V(X_t) = \frac{\phi^k \sigma^2}{1-\phi^2}.$$

p.4 bottom

Eg. MA (2)

From p. 3 bottom, we have $\rho(k) = \begin{cases} \dfrac{\sum_{i=0}^{m-k} \beta_i \beta_{i+k}}{\sum_{i=0}^{m} \beta_i^2}, & k = 1, 2, \ldots, m; \\ 0, & k > m. \end{cases}$

Now, $m = 2$

$$\rho(k) = \frac{\sum_{i=0}^{m-k} \beta_i \beta_{i+k}}{\sum_{i=0}^{m} \beta_i^2}$$

$$\rho(1) = \frac{\sum_{i=0}^{1} \beta_i \beta_{i+1}}{\sum_{i=0}^{2} \beta_i^2} = \frac{\beta_0\beta_1 + \beta_1\beta_2}{\beta_0^2 + \beta_1^2 + \beta_2^2} = \frac{\beta_1 + \beta_1\beta_2}{1 + \beta_1^2 + \beta_2^2}. \quad (\text{for } \beta_0 = 1)$$

$$\rho(2) = \frac{\sum_{i=0}^{0} \beta_i \beta_{i+2}}{\sum_{i=0}^{2} \beta_i^2} = \frac{\beta_0\beta_2}{1 + \beta_1^2 + \beta_2^2} = \frac{\beta_2}{1 + \beta_1^2 + \beta_2^2}.$$

p.5 Example: For an invertible MA(2) process

$$Y_t = \varepsilon_t - 0.2\,\varepsilon_{t-1} + \theta_2\,\varepsilon_{t-2},$$

you are given that $\rho(1) = -0.1$. Determine $\rho(2)$.

From the information given, $m = 2$, $\theta_1 = -0.2\ (= \beta_1)$, $\theta_0 = 1\ (= \beta_0)$.

Using the formulas above,

$$\rho(1) = \rho_1 = -0.1 = \frac{\sum_{i=0}^{2-1} \theta_i \theta_{i+1}}{\sum_{i=0}^{2} \theta_0^2} = \frac{\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2} = \frac{-0.2 - 0.2\theta_2}{1 + (-0.2)^2 + \theta_2^2}$$

$$\Rightarrow \quad -0.1 = \frac{-0.2 - 0.2\theta_2}{1 + 0.04 + \theta_2^2} \quad \Rightarrow \quad 0.1 = \frac{0.2 + 0.2\theta_2}{1 + 0.04 + \theta_2^2}$$

$$\Rightarrow \quad 0.1 + 0.004 + 0.1\theta_2^2 = 0.2 + 0.2\theta_2 \qquad (\times 10)$$

$$\Rightarrow \quad 1 + 0.04 + \theta_2^2 - 2 - 2\theta_2 = 0$$

$$\Rightarrow \quad \theta_2^2 - 2\theta_2 - 0.96 = 0 = (\theta_2 - 2.4)(\theta_2 + 0.4)$$

$$\therefore \quad \theta_2 = \begin{cases} -0.4 \\ 2.4 \quad (\text{rejected, since } 2.4 > 1) \end{cases}$$

$$\rho(2) = \rho_2 = \frac{\sum_{i=0}^{2-2} \theta_i \theta_{i+2}}{1 + \theta_1^2 + \theta_2^2} = \frac{\theta_0\theta_2}{1 + \theta_1^2 + \theta_2^2} = \frac{-0.4}{1 + (-0.2)^2 + (-0.4)^2} = \frac{-0.4}{1 + 0.04 + 0.16} = \frac{-0.4}{1.2} = -\frac{1}{3}$$

P.5 3.3.4 Random Walk

$X_t = X_{t-1} + Z_t$, where Z_t white noise

Z_t and X_{t-k} are uncorrelated, $k>0$.

$E(X_t) = E(X_{t-1}) + E(Z_t)$

$\mu = \mu + 0$

$\Rightarrow$ $E(X_t)$ does not depend on t, hence mean stationary.

However,

$V(X_t) = V(X_{t-1}) + V(Z_t) = V(X_{t-1}) + \sigma^2$

$V(X_{t-1}) = V(X_{t-2}) + V(Z_{t-1}) = V(X_{t-2}) + \sigma^2$

$V(X_{t-2}) = V(X_{t-3}) + V(Z_{t-2}) = V(X_{t-3}) + \sigma^2$

by substitution, then

$V(X_t) = V(X_{t-3}) + 3\sigma^2$

after t steps

$\Rightarrow V(X_t) = V(X_0) + t\sigma^2$, which depends on t, hence the process is not stationary.

But, $Y_t = \nabla X_t = X_t - X_{t-1} = Z_t$ is stationary

because the mean, variance and ACF of Y_t all do not depend on t.

$E(Y_t) = E(Z_t) = 0$

$V(Y_t) = V(Z_t) = \sigma^2$

$Cov(Y_t, Y_{t+k}) = Cov(Z_t, Z_{t+k}) = E(Z_t Z_{t+k}) = \gamma(k)$

all do not depend on t.

3.4.1

Infinite order

$$X_t = \alpha X_{t-1} + Z_t \qquad AR(1)$$

$$X_t - \alpha X_{t-1} = Z_t$$

$$(1-\alpha B) X_t = Z_t$$

$$X_t = (1-\alpha B)^{-1} Z_t$$

$$= (1 + \alpha B + \alpha^2 B^2 + \cdots) Z_t$$

$$= \sum_{i=0}^{\infty} \alpha^i B^i Z_t$$

$$= \sum_{i=0}^{\infty} \alpha^i Z_{t-i} \qquad MA(\infty)$$

$$\Rightarrow AR(1) \equiv MA(\infty)$$

Finite order

$$X_t = \Theta(B) Z_t = (\beta_0 + \beta_1 B + \beta_2 B^2 + \cdots + \beta_m B^m) Z_t \,, \quad \beta_0 = 1,$$

$$= \sum_{i=0}^{m} \beta_i Z_{t-i} \qquad MA(m)$$

For an $MA(1)$ process, $X_t = Z_t + \beta Z_{t-1}$ with $|\beta| < 1$. (3.4)

$$\rho(0) = \frac{\sum_{i=0}^{1} \beta_i \beta_i}{\sum_{i=0}^{1} \beta_i^2} = 1, \quad \text{where } \beta_i = \beta \quad i = 1$$
$$= 0 \quad i > 1$$

$$\rho(1) = \frac{\sum_{i=0}^{0} \beta_i \beta_{i+1}}{\sum_{i=0}^{1} \beta_i^2} = \frac{\beta_0 \beta_1}{1+\beta_1^2} = \frac{\beta}{1+\beta^2}.$$

$$\rho(k) = 0, \text{ for } k > 1.$$

p.7 Consider the process

$$X_t = Z_t + \frac{1}{\beta} Z_{t-1} \qquad MA(1)$$

$$= \left(1 + \frac{B}{\beta}\right) Z_t \qquad (3.5) \qquad \left|\frac{1}{\beta}\right| > 0, \text{ i.e., } |\beta| < 1.$$

For (3.4), $X_t = (1 + \beta B) Z_t$

$$\Rightarrow Z_t = (1+\beta B)^{-1} X_t = (1 - \beta B + \beta^2 B^2 - \beta^3 B^3 + \beta^4 B^4 - \cdots) X_t$$

$$= X_t - \beta X_{t-1} + \beta^2 X_{t-2} - \beta^3 X_{t-3} + \beta^4 X_{t-4} - \cdots$$

Whence the coefficients on the RHS is decaying and convergent.

p.8 For (3.5), $X_t = \left(1 + \frac{B}{\beta}\right) Z_t$

$$\Rightarrow Z_t = \left(1 + \frac{B}{\beta}\right)^{-1} X_t = \left(1 - \frac{B}{\beta} + \frac{B^2}{\beta^2} - \frac{B^3}{\beta^3} + \frac{B^4}{\beta^4} - \cdots\right) X_t$$

$$= X_t - \frac{1}{\beta} X_{t-1} + \frac{1}{\beta^2} X_{t-2} - \frac{1}{\beta^3} X_{t-3} + \frac{1}{\beta^4} X_{t-4} - \cdots$$

(3.5^*)

The coefficients on the RHS are not decaying and not convergent.

$\Rightarrow$ $MA(1) \equiv AR(\infty)$

$$X_t = \Theta(B)\, Z_t$$

$$Z_t = [\Theta(B)]^{-1} X_t \qquad (3.5^*)$$

If (3.5^*) is uncertain, then the time series X_t is not invertible. However, a non-invertible time series may also have problems in statistical estimation.

3.4.2 Finite order autoregressive process AR(m)

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_m X_{t-m} + Z_t \qquad (3.6)$$

$$\begin{aligned} Z_t &= X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} - \cdots - \alpha_m X_{t-m} \\ &= X_t - \alpha_1 B X_t - \alpha_2 B^2 X_t - \cdots - \alpha_m B^m X_t \\ &= (1 - \alpha_1 B - \alpha_2 B^2 - \cdots - \alpha_m B^m) X_t \\ &= \Phi(B) X_t \end{aligned}$$

Assume stationary and let $E(X_t) = 0$, then consider the multiple of (3.6) by X_{t-k} and take expectation.

$$X_t X_{t-k} = \alpha_1 X_{t-1} X_{t-k} + \alpha_2 X_{t-2} X_{t-k} + \cdots + \alpha_m X_{t-m} X_{t-k} + Z_t X_{t-k}$$

$$\begin{aligned} \gamma(k) &= E(X_t X_{t-k}) \\ &= \alpha_1 E(X_{t-1} X_{t-k}) + \alpha_2 E(X_{t-2} X_{t-k}) + \cdots + \alpha_m E(X_{t-m} X_{t-k}) + E(Z_t) E(X_{t-k}) \\ &= \alpha_1 \gamma(k-1) + \alpha_2 \gamma(k-2) + \cdots + \alpha_m \gamma(k-m), \quad k = 1, 2, \ldots, m. \end{aligned}$$

Hence

$$\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2) + \cdots + \alpha_m \rho(k-m), \quad k = 1, 2, \ldots, m. \qquad (3.7)$$

Yule-Walker equation

In principle, (3.7) can be solved to give $\rho(k)$ in terms of α_i's and vice versa.

p.9 (3.7) can be written in matrix form

$$\begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(m) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) & \rho(2) & \cdots & \rho(m-1) \\ \rho(1) & 1 & \rho(3) & \cdots & \rho(m-2) \\ \rho(2) & \rho(3) & 1 & \cdots & \rho(m-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho(m-1) & \rho(m-2) & \rho(m-3) & \cdots & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix}$$

$\rho(k) = \rho(-k)$ and $\rho(0) = 1$ for stationary $\{X_t\}$,

i.e., $\rho_{m\times 1} = R_{m\times m}\,\alpha_{m\times 1}$, and provided R^{-1} exists.

[Note that $R_{m\times m}$ is symmetric.] Hence we have

$$\alpha_{m\times 1} = R^{-1}_{m\times m}\,\rho_{m\times 1}$$

i.e., given the 1st m autocorrelations, or estimates of the correlation parameters, then the mth order autoregressive process may be obtained or estimated.

Theorem: A necessary and sufficient condition for the stationarity of the autoregressive process

$$\Phi(B)\,X_t = Z_t$$

where $\Phi(B)$ is a polynomial in B of order m, is that the roots of the equation

$$\Phi(B) = 0$$

should all lie outside the unit circle.

Proof: We first show the AR(2) case and then generalize.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

The solution to the homogeneous difference equation

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} = 0$$

depends on the roots r_1, r_2 of the equation

$$1 - \phi_1 B - \phi_2 B^2 = 0$$

which, we shall assume, are distinct although they may be complex (i.e., in terms of $i = \sqrt{-1}$). The complementary function is of the form

$$X_t = k_1 \left(\frac{1}{r_1}\right)^t + k_2 \left(\frac{1}{r_2}\right)^t,$$

where k_1 and k_2 are constants depending on initial conditions. The particular solution is the infinite MA representation

$$X_t = [\phi(B)]^{-1} \varepsilon_t = \psi(B) \varepsilon_t.$$

The general solution to the AR difference equation is therefore

$$X_t = k_1 \left(\frac{1}{r_1}\right)^t + k_2 \left(\frac{1}{r_2}\right)^t + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.$$

Similar to p.2 (3.3).

For the model to have a constant mean, the complementary function must decay for large value of t, so we require

$\left|\frac{1}{r_1}\right| < 1$ and $\left|\frac{1}{r_2}\right| < 1$. ($|r_1| > 1$ and $|r_2| > 1$)

The AR polynomial takes the form

$$\phi(B) = \left(1 - \frac{1}{r_1}B\right)\left(1 - \frac{1}{r_2}B\right),$$

where r_1 and r_2 are the roots. A particular solution is

$$X_t = \psi(B)\varepsilon_t = [\phi(B)]^{-1}\varepsilon_t,$$

so using partial fractions we can write

$$[\phi(B)]^{-1} = \frac{1}{\left(1 - \frac{1}{r_1}B\right)\left(1 - \frac{1}{r_2}B\right)}$$

$$= \frac{a}{1 - \frac{1}{r_1}B} + \frac{b}{1 - \frac{1}{r_2}B}$$

where a and b are constants. Each of the terms in $[\phi(B)]^{-1}$ can be expanded – for instance, the first term is

$$a\left[1 + \frac{1}{r_1}B + \left(\frac{1}{r_1}B\right)^2 + \cdots\right]$$

and the second term is

$$b\left[1 + \frac{1}{r_2}B + \left(\frac{1}{r_2}B\right)^2 + \cdots\right]$$

so if the variance of $X_t = \psi(B)\varepsilon_t = [\phi(B)]^{-1}\varepsilon_t$ is to converge we require that $\left|\frac{1}{r_1}\right| < 1$ and $\left|\frac{1}{r_2}\right| < 1$

$\Rightarrow |r_1| > 1$ and $|r_2| > 1$.

Hence the condition for an AR(2) model to be stationary is that both the roots r_1 and r_2 of the AR polynomial are outside the unit circle, i.e., $|r_1| > 1$ and $|r_2| > 1$.

Similar logic can be applied to the general AR(p) model. Assuming that the roots of the corresponding AR polynomial

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

are distinct and denoting these $r_1, r_2, \ldots, r_p$. Hence the complementary function is

$$k_1 \left(\frac{1}{r_1}\right)^t + k_2 \left(\frac{1}{r_2}\right)^t + \cdots + k_p \left(\frac{1}{r_p}\right)^t.$$

Thus similar to the AR(2) argument if the roots are all outside the unit circle. The complementary function will decay to zero for large t and the model will be stationary. Conversely, i.e., the model is stationary and thus has finite variance. Let $\lambda_i = \frac{1}{r_i}$, then

$$X_t = [\phi(B)]^{-1} \varepsilon_t = \left[\prod_{i=1}^{p} (1 - \lambda_i B)\right]^{-1} \varepsilon_t.$$

Assume again the roots are distinct, then

$$X_t = \left\{\sum_{i=1}^{p} \frac{a_i}{1 - \lambda_i B}\right\} \varepsilon_t = \sum_{i=1}^{p} \sum_{j=1}^{\infty} a_i (\lambda_i B)^j \varepsilon_t = \sum_{i=1}^{p} \sum_{j=0}^{\infty} a_i \lambda_i^j \varepsilon_{t-j}$$

So $E(X_t) = E\left(\sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^j \varepsilon_{t-j}\right)$

$$= \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^j E(\varepsilon_{t-j}) = 0$$

and $V(X_t) = V\left(\sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^j \varepsilon_{t-j}\right)$

$$= \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i^2 \lambda_i^{2j} V(\varepsilon_{t-j})$$

$$= \sigma^2 \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i^2 \lambda_i^{2j}.$$

$V(X_t)$ is finite implies the absolute values of all λ_i's are less than 1.

Eg. An AR(2) process : $X_t = 0.7X_{t-1} - 0.1X_{t-2} + Z_t$

$\{Z_t\} \sim NID(0, \sigma^2)$

Then $(1 - 0.7B + 0.1B^2)X_t = Z_t$.

The roots of the backshift polynomial in B, i.e.,

$$1 - 0.7B + 0.1B^2 = (1 - 0.2B)(1 - 0.5B) = 0$$

$$B = \begin{cases} 5 \\ 2 \end{cases}$$

all greater than 1, which means that the roots are all lied outside the unit circle. Thus the AR(2) process is stationary.

3.4.3 Mixed processes (ARMA processes)

The equation:

$$\alpha_0 X_t - \alpha_1 X_{t-1} - \cdots - \alpha_p X_{t-p} = Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q}$$

$\{Z_t\}$ is white noise and conventionally $\alpha_0 = 1$ defines a mixed autoregressive moving average process of order (p, q), i.e., ARMA(p, q).

In operator notation, we have

$$\Phi(B) X_t = \Theta(B) Z_t .$$

Autocorrelation functions are available in principal via the equation

$$X_t = \frac{\Theta(B)}{\Phi(B)} Z_t$$

which expresses X_t as an MA(∞) process.

N.B. $[\Phi(B)]^{-1} = \frac{1}{\Phi(B)}$ is an infinite series.

Or the other round,

$$Z_t = \frac{\Phi(B)}{\Theta(B)} X_t$$

which expresses Z_t as an AR(∞) process.

N.B. $[\Theta(B)]^{-1} = \frac{1}{\Theta(B)}$ is an infinite series.

Eg. An ARMA(2, 1) process

$$X_t - 0.75X_{t-1} + 0.5625X_{t-2} = Z_t + 1.25Z_{t-1}$$

$\{Z_t\} \sim NID(0, \sigma^2)$.

$$\phi(B) = 1 - 0.75B + 0.5625B^2 = 1 - \frac{3}{4}B + \frac{9}{16}B^2 = 0$$

$$\Rightarrow 16 - 12B + 9B^2 = 0$$

$$\Rightarrow 9B^2 - 12B + 16 = 0$$

$$B = \frac{12 \pm \sqrt{(-12)^2 - 4(9)(16)}}{2(9)} = \frac{12 \pm \sqrt{-432}}{18}$$

$$= \frac{12 \pm 12\sqrt{3}\,i}{18}$$

$$= \frac{2 \pm 2\sqrt{3}\,i}{3} > 1 \quad \text{(outside the unit circle)}$$

$$\theta(B) = 1 + 1.25B$$

$$= 1 + \frac{5}{4}B = 0 \Rightarrow B = -\frac{4}{5} = -0.8 \quad \text{with } |-0.8| < 1.$$

Hence the process is stationary for $\phi(B)$ but the process is not invertible for $\theta(B)$.

In practice, many sets of data can reasonably be modelled by either a high order MA process or a low order AR process. If so, we positively refer the version with fewer parameters. Similarly, an ARMA(p, q) process of lower order can often be made to fit data as either high order AR or MA process.

3.4.4 Low order models

Experience suggest that ARMA(p,q) process with $p+q \leq 2$ constitute an extremely useful class of parsimonious models for stationary time series data. We can summerize main features of AR(2), MA(2) and ARMA(1,1) processes.

AR(2) Process

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + Z_t$$

$$\Rightarrow \quad X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} = Z_t$$

$$\Rightarrow \quad (1 - \alpha_1 B - \alpha_2 B^2) X_t = Z_t$$

then $$\Phi(B) X_t = Z_t$$

and $$\Phi(B) = 1 - \alpha_1 B - \alpha_2 B^2$$

then $$1 - \alpha_1 B - \alpha_2 B^2 = 0 = \alpha_2 B^2 + \alpha_1 B - 1$$

$$\Rightarrow \quad B = \frac{-\alpha_1 \pm \sqrt{(\alpha_1)^2 - 4\alpha_2(-1)}}{2\alpha_2}$$

$$= \frac{-\alpha_1 \pm \sqrt{\alpha_1^2 + 4\alpha_2}}{2\alpha_2}$$

For stationarity, it requires that $|B| > 1$.

The Yule-Walker equations become (refer to p.8 (3.7) with $m = 2$)

$$\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2) \qquad k \geq 0 \quad (3.8)$$

$\rho(0) = \alpha_1 \rho(1) + \alpha_2 \rho(2)$ N.B. $\rho(k) = \rho(-k)$ and $\rho(0) = 1$

$\rho(1) = \alpha_1 \rho(0) + \alpha_2 \rho(1)$

$\rho(1) = \alpha_1 + \alpha_2 \rho(1) \Rightarrow (1-\alpha_2)\rho(1) = \alpha_1 \Rightarrow \rho(1) = \frac{\alpha_1}{1-\alpha_2}$

$\rho(2) = \alpha_1 \rho(1) + \alpha_2 \rho(0) = \alpha_1 \cdot \frac{\alpha_1}{1-\alpha_2} + \alpha_2 = \frac{\alpha_1^2}{1-\alpha_2} + \alpha_2.$

Note that the constraints on α_1, α_2 imply the constraints on $\rho(1)$ and $\rho(2)$. Alternatively, since (3.8) is a linear, 2nd order difference equation, it can be solved explicitly. In fact, for (3.8), i.e., $\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2)$, $k \geq 0$, the general solution is

$$\rho(k) = A_1 \lambda_1^k + A_2 \lambda_2^k$$

where λ_1, λ_2 are the roots of $\lambda^2 - \alpha_1 \lambda - \alpha_2 = 0$.

(see p. 9)

That is,

$$\Phi(B) = 1 - \alpha_1 B + \alpha_2 B^2 = 0$$

let $B = \frac{1}{\lambda}$, then

$$1 - \frac{\alpha_1}{\lambda} - \frac{\alpha_2}{\lambda^2} = 0 \Rightarrow \lambda^2 - \alpha_1 \lambda - \alpha_2 = 0.$$

Values of A_1 and A_2 can be obtained from using the initial values of $\rho(0)$ and $\rho(1)$. Thus we have 3 distinct forms of solution:

(i) $\alpha_1^2 + 4\alpha_2 > 0 \Rightarrow$ mixtures of 2 exponentially decaying functions for $\rho(k)$;

(ii) $\alpha_1^2 + 4\alpha_2 < 0 \Rightarrow \lambda = a e^{\pm i\theta}$, $\rho(k)$ is a damped harmonic;

(iii) $\alpha_1^2 + 4\alpha_2 = 0 \Rightarrow \rho(k) = (A_1 + A_2 k)\lambda^k$, i.e., $\lambda_1 = \lambda_2 = \lambda$, an exponential decay.

MA(2) Process

$$X_t = Z_t + \beta_1 Z_{t-1} + \beta_2 Z_{t-2} = (1 + \beta_1 B + \beta_2 B^2) Z_t$$

$$\Rightarrow X_t = \Theta(B) Z_t .$$

For invertibility, require roots of $1 + \beta_1 B + \beta_2 B^2 = 0$ to be outside the unit circle, i.e.,

$$\left| \frac{-\beta_1 \pm \sqrt{\beta_1^2 - 4\beta_2}}{2\beta_2} \right| > 1$$

which implies the same restriction on $(-\beta_1, -\beta_2)$ as for (α_1, α_2) in AR(2) process. Again restrictions on $-\beta_1$ and $-\beta_2$ imply restrictions on $\rho(1)$ and $\rho(2)$.

ARMA(1, 1) Process

$$X_t - \alpha X_{t-1} = Z_t + \beta Z_{t-1}$$

$$\Rightarrow (1 - \alpha B) X_t = (1 + \beta B) Z_t$$

$$\Rightarrow X_t = \frac{(1+\beta B)}{1 - \alpha B} Z_t = (1+\beta B)(1 + \alpha B + \alpha^2 B^2 + \cdots) Z_t$$

Thus $|\alpha| < 1$ and $|\beta| < 1$ are together sufficient to give a stationary and invertible process $\{X_t\}$. Then

$$X_t = (1+\beta B) \sum_{i=0}^{\infty} \alpha^i B^i Z_t = \left[\begin{matrix} 1 + \alpha B + \alpha^2 B^2 + \alpha^3 B^3 + \cdots \\ + \quad \beta B + \alpha\beta B^2 + \alpha^2 \beta B^3 + \cdots \end{matrix} \right] Z_t$$

$$= \left[1 + (\alpha+\beta) B + \alpha(\alpha+\beta) B^2 + \alpha^2 (\alpha+\beta) B^3 + \cdots \right] Z_t$$

$$= \left\{ 1 + \sum_{i=1}^{\infty} \alpha^{i-1} (\alpha+\beta) B^i \right\} Z_t$$

$$X_t = \sum_{i=0}^{\infty} b_i Z_{t-i}$$

where b_i's are respectively, for $i \geq 1$;

$$b_0 = 1$$
$$b_1 = \alpha + \beta$$
$$b_2 = \alpha(\alpha+\beta)$$
$$b_3 = \alpha^2(\alpha+\beta)$$
$$\vdots$$
$$b_i = \alpha^{i-1}(\alpha+\beta)$$

Using the results of section 3.3.3 $\rho(k) = \dfrac{\sum_{i=0}^{\infty} b_i b_{i+k}}{\sum_{i=0}^{\infty} b_i^2}$ (with $m \to \infty$)

(please see p. 3 bottom, the β_i's are changed to b_i's.)

$$\sum_{i=0}^{\infty} b_i^2 = b_0^2 + b_1^2 + b_2^2 + \cdots + b_i^2 + \cdots$$
$$= 1 + (\alpha+\beta)^2 + \alpha^2(\alpha+\beta)^2 + \cdots + \alpha^{2i-2}(\alpha+\beta)^2 + \cdots$$
$$= 1 + (\alpha+\beta)^2\left[1 + \alpha^2 + \alpha^4 + \cdots + \alpha^{2i-2} + \cdots\right]$$
$$= 1 + \frac{(\alpha+\beta)^2}{1-\alpha^2}$$

$$\sum_{i=0}^{\infty} b_i b_{i+k} = b_0 b_k + b_1 b_{k+1} + b_2 b_{k+2} + \cdots = b_k + b_1 b_{k+1} + b_2 b_{k+2} + \cdots$$
$$= \alpha^{k-1}(\alpha+\beta) + (\alpha+\beta)\alpha^k(\alpha+\beta) + \alpha(\alpha+\beta)\alpha^{k+1}(\alpha+\beta) + \cdots$$
$$= \alpha^{k-1}(\alpha+\beta) + (\alpha+\beta)^2\alpha^k + (\alpha+\beta)^2\alpha^{k+2} + \cdots$$
$$= \alpha^{k-1}(\alpha+\beta) + (\alpha+\beta)^2\alpha^k \frac{1}{1-\alpha^2}$$

$$\rho(k) = \frac{\alpha^{k-1}(\alpha+\beta) + \frac{(\alpha+\beta)^2\alpha^k}{1-\alpha^2}}{1 + \frac{(\alpha+\beta)^2}{1-\alpha^2}} = \frac{\alpha^{k-1}(\alpha+\beta)\left[1 + \frac{\alpha(\alpha+\beta)}{1-\alpha^2}\right]}{1 + \frac{(\alpha+\beta)^2}{1-\alpha^2}}$$

$$= \frac{\alpha^{k-1}(\alpha+\beta)\left[1-\alpha^2+\alpha^2+\alpha\beta\right]/(1-\alpha^2)}{(1-\alpha^2+\alpha^2+2\alpha\beta+\beta^2)/(1-\alpha^2)} = \frac{\alpha^{k-1}(\alpha+\beta)(1+\alpha\beta)}{1+2\alpha\beta+\beta^2}$$

p.14 Example ARMA(2, 1) process

$$y_t = 0.3\, y_{t-1} + 0.2\, y_{t-2} + \varepsilon_t + \theta\, \varepsilon_{t-1}$$

$\alpha_1 = 0.3$, $\alpha_2 = 0.2$ and given $\rho(1) = 0.8$, $\rho(2) = 0.44$.

Find $\rho(4)$.

Making use of Yule-Walker equations

$$\begin{aligned}\rho(4) &= 0.3\,\rho(3) + 0.2\,\rho(2) \\ &= 0.3\,\rho(3) + 0.2\,(0.44) \\ &= 0.3\,\rho(3) + 0.088\end{aligned}$$

$$\begin{aligned}\rho(3) &= 0.3\,\rho(2) + 0.2\,\rho(1) \\ &= 0.3\,(0.44) + 0.2\,(0.8) \\ &= 0.292\end{aligned}$$

$$\therefore \rho(4) = 0.3\,(0.292) + 0.088 = 0.1756.$$

3.5 Integrated process ARIMA(p, d, q)

Recall the Markov process $X_t = X_{t-1} + Z_t$, which is nonstationary but can be transformed to a stationary process by differencing.

More generally, consider the process

$$\Omega(B)\, X_t = \Theta(B)\, Z_t \qquad (3.9)$$

where $\{Z_t\}$ is white noise and $\Omega(B) = \Phi(B)(1-B)^d$.

$\Phi(B)$ and $\Theta(B)$ are polynomials of degree p and q, respectively, all of their roots are outside the unit circle. Note that, so we can write (3.9) as

$$\Phi(B)\{\nabla^d X_t\} = \Theta(B)\, Z_t$$

$$\Rightarrow \quad \Phi(B)(1-B)^d X_t = \Theta(B)\, Z_t$$

[d is the seasonal operator.]

i.e., the dth difference of $\{X_t\}$ is ARMA(p, q), we call (3.9) an autoregressive integrated moving average process or $\{X_t\}$ is an ARIMA(p, d, q) process. $\{X_t\}$ is nonstationary, so $\rho(k)$ does not exist. However, $\hat{\rho}(k)$ can be useful in identifying ARIMA process. Typically slow decay from $\hat{\rho}(1) \approx 1$ suggests an integrated process a plausible model.

Warning: over difference an already stationary series will after produce an undesirable increase in variability.

3.6 Seasonal models

The seasonal difference operator

$$\nabla_s X_t = X_t - X_{t-s} = (1 - B^s) X_t$$

Can be used to eliminate seasonal nonstationarity.

$1 - B^s$ is the seasonal operator

seasonally => spring, summer, autumn and winter
then $s = 4$.

monthly => January, February, March, ..., December
then $s = 12$.

Formally, consider ARIMA models of the form

$$\Phi(B)(1-B)^{d_1}(1-B^s)^{d_2} X_t = \Theta(B) Z_t$$

$$\text{ARIMA}(p, d_1, q) \times (0, d_2, 0)_s$$

$$\Phi(B)(1-B)^{d} X_t = \Theta(B) Z_t$$

ARIMA (p, d, q) without seasonal difference.